

Real time technologies for Realistic Digital Humans: facial performance and hair simulation

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1 OVERVIEW



We have identified the world's most extensive parameter space for the human head based on over 4TB of 4D data acquired from multiple actors. Ziva's proprietary machine-learning processes can apply this data set to any number of secondary 3D heads, enabling them all to perform novel facial expressions in real-time while preserving volume, achieving accurate secondary dynamics, and staying within the natural range of human movement. Facial performances can then be augmented and tailored with Ziva expressions controls, solving the costly limitations of scalability, realism, artist control, and speed.

In this presentation, we discuss and demonstrate how this innovation can improve the overall quality of RT3D faces for linear and real-time 3D productions while simplifying and accelerating the entire asset production and digital casting workflows, enabling mass production of high-performance real-time content. We will then illustrate how performance capture can be decoupled from asset production by showing a single performance being applied to multiple faces of varying proportions. This will highlight the flexibility, state-of-the-art quality, and extensive time-saving made possible by actor-agnostic / actor-nonspecific performance capture.

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We will additionally highlight a newly integrated hair solution for authoring / importing / simulating / rendering strand-based hair in Unity. Built from the ground up with Unity users in mind, and evolved and hardened during the production of *Enemies*, the hair system is applicable to realistic digital humans and much more stylized content and games. Using a fast and flexible GPU-based solver that works on both strand- and volume-information, the system enables users to interactively set up 'Hair Instances' and interact with those instances as they are simulated and rendered in real time.